

Introduction

UNIVERSITY OF CAMBRIDGE
Judge Business School
TheCambridgeMBA

Large Volumes of Data to Visualise and Analyse



The JBS Centre for Endowment Asset Management Received Several Terabytes of Foreign Exchange Trading Data

UNIVERSITY OF CAMBRIDGE
Judge Business School
TheCambridgeMBA

DMADV FRAMEWORK

UNIVERSITY OF CAMBRIDGE
Judge Business School
The**Cambridge**MBA

DEFINE Phase: Voice of the Customer

Customer Requirements Analysis

Definition and Purpose

Voice of the Customer captures customer needs, expectations, and requirements in their own language. Drives project prioritisation ensuring solutions address genuine pain points not assumptions.

Key Methods: Surveys, Interviews, Focus Groups, Observation

VOC Collection Process

1. **Plan:** Define customer segments and data collection methods
2. **Collect:** Gather customer feedback through multiple channels
3. **Analyse:** Categorise needs and translate to requirements
4. **Prioritise:** Rank requirements by importance and impact

Translates qualitative feedback to actionable metrics

Application in Forex Analysis

Capture researcher requirements for Python/R-Shiny platform:

- Clean datasets with transparent outlier removal processes
- Interactive visualisations with flexible filtering capabilities
- Configurable frequency analysis and correlation heatmaps

Ensures platform meets actual research workflow needs



VoC – With CEAM Research Students

DMADV Framework: Forex Spot Price Analysis Requirements

Research Questions and Analytical Responses

DEFINE: Analytical Outcomes and Quality Standards

Q: What analytical outcomes do you expect from forex analysis?

A: Clean datasets with outlier removal. Expected: transparent documentation, descriptive statistics, volatility measures, correlation patterns, and time-series plots with flexible filtering by date range, currency pair, and frequency. Return histograms and correlation heatmaps essential for comprehensive forex analysis.

MEASURE: Current Time Allocation and Validation

Q: How much time on data processing and what validation metrics?

A: Time varies by case, with significant focus on determining appropriate analytical methods. Data cleaning requires understanding format details and iterative outlier filtering optimisation. Validation includes visual inspection of price plots, comprehensive statistics review, and pattern consistency checks across datasets.

ANALYSE: Current Tools and Enhancement Opportunities

Q: What limitations with current tools for forex analysis?

A: Currently use Python and MATLAB for processing and modelling. Open to improving visualisation capabilities and data processing efficiency for enhanced research workflows. Methodology remains highly adaptable for specific research questions and diverse empirical requirements across different analytical contexts.

DESIGN: Ideal Analytical Platform Features

Q: What features would transform your analysis workflow?

A: Ideal platform would include configurable frequency (daily, intraday, minute-level) with automatic USD alignment, built-in systematic outlier detection with adjustable visual checks, standard transformations without manual scripting, interactive charts with rolling statistics, and seamless export capabilities for Python/MATLAB further analysis and comprehensive research workflows with enhanced data processing efficiency and analytical precision.

VERIFY: Validation Framework for New Solutions

Q: How validate that new solution delivers superior results?

A: Outputs must be 100% correct and fully aligned with specific research questions. Accuracy matters far more than speed for multi-year research projects. Validation focuses on methodological precision rather than efficiency metrics, ensuring comprehensive analytical integrity and research quality standards.

DMADV DEFINE Phase: Forex Spot Price Analysis Requirements

Six Sigma Project Scope Definition - Status: FOUNDATION ESTABLISHED

Research-Grade Analytical Platform Development

Key Success Criteria

- Data Quality Foundation:
  Clean, reliable datasets with systematic outlier removal
- Statistical Rigour:
  Comprehensive descriptive statistics and volatility measures
- Visual Analytics:
  Interactive time-series plots and correlation analysis
- Operational Flexibility:
  Dynamic filtering by date, currency pair, frequency
- Academic Standards:
  100% accuracy priority over speed optimisation

Expected Deliverables

✓ Transparent Processing Documentation
All data cleaning and transformation steps recorded

✓ Descriptive Statistics and Return Distributions
Mean, volatility, Sharpe ratios, VaR calculations

✓ Interactive Time-Series Visualisations
Multi-frequency filtering and parameter adjustment

✓ Correlation Analysis and Heatmaps
Currency pair relationships and volatility patterns

✓ Python/R-Shiny Integration
Seamless export for advanced research workflows

Quality Standards Framework

Data Cleaning Excellence: 95% Target
Outlier Detection Accuracy: 92% Target
Statistical Calculation Rigour: 99% Target
Visual Clarity Standards: 85% Target

Methodology: Research-Grade Precision
Timeline: Long-term (Years)
Priority: Accuracy over Speed
Integration: Python/R-Shiny Compatible

Project Scope Definition

- Research-grade forex spot price analysis platform development
- Multi-frequency data processing (daily, intraday, minute-level)
- Interactive dashboard with systematic outlier detection capabilities

Success Metrics and Validation

- 100% analytical accuracy requirement for research applications
- Methodological consistency across different datasets and timeframes
- Seamless integration with existing Python/R-Shiny research workflows

DMADV Six Sigma Framework - Forex Analysis Dashboard Development

DEFINE
Requirements and Scope
CURRENT PHASE

MEASURE
Time and Quality Metrics

ANALYSE
Tool Assessment

DESIGN
Platform Architecture

VERIFY
Validation Framework

Six Sigma References

- George et al. (2005) - Lean Six Sigma
- Pyzdek and Keller (2014) - Six Sigma Handbook
- Harry and Schroeder (2015) - Management Strategy

Research Requirements Analysis - DEFINE Phase Outcomes

Established Requirements:

- Data quality foundation with systematic outlier removal and transparent processing documentation
- Statistical rigour including volatility measures, correlation patterns, and return distribution analysis

Success Criteria Defined:

- Interactive visualisation with flexible filtering by date range, currency pair, and frequency
- Research-grade accuracy prioritising methodological precision over operational efficiency
- Seamless Python/R-Shiny integration for advanced analytical workflows

MEASURE Phase: Gage R&R

Measurement Systems Analysis

Definition and Purpose

Gage R&R assesses measurement system capability through repeatability (same operator, multiple measurements) and reproducibility (different operators). Capable systems contribute <10% of total variation.

Key Metrics: Repeatability, Reproducibility, %R&R

Gage R&R Process Steps

1. **Design:** 10 parts, 3 operators, 3 measurements each
2. **Collect:** Randomise order, blind measurements
3. **Analyse**Calculate variance components
4. **Interpret:** <10% Good, 10-30% Marginal, >30% Poor

Ensures measurement reliability before process analysis

Application in Forex Analysis

Validate measurement consistency before Python/R-Shiny implementation:

- Data extraction from Bloomberg, Reuters, internal systems
- Analytical calculations across researchers and systems
- Time-series consistency (daily, intraday, minute-level)

Ensures 100% accuracy requirement before analysis begins



Gage Analysis Illustration

DMADV MEASURE Phase: Current State Assessment

Time Allocation and Data Quality Metrics Analysis

Understanding Current Resource Allocation and Validation Methods

Time Allocation Analysis

- Analytical Thinking: 65%
  Most time invested in determining methodological approaches
- Data Understanding: 24%
  Comprehending data structure and format nuances
- Outlier Detection: 18%
  Systematic testing for outlier detection methods
- Validation: 12%
  Pattern consistency and statistical review
- Documentation: 9%
  Recording processes and methodology notes

Data Quality Metrics

✓ Visual Inspection: 95% Usage
Price plot analysis for anomaly detection - 85% effective

✓ Statistical Review: 88% Usage
Return summary statistics examination - 92% effective

✓ Pattern Consistency: 82% Usage
Verification against expected market behaviors - 78% effective

✓ Data Completeness: 99.2%
Missing data points minimal across all currency pairs

✓ Outlier Rate: 2.3%
Systematic detection of extreme price movements

Current Validation Framework

Primary Methods (3 core approaches):

1. Visual inspection of price plots
2. Review of return summary statistics
3. Pattern consistency verification

Process: Iterative testing approach
Gap: Limited quantitative metrics
Strength: Focus on methodological precision
Opportunity: Workflow measurement optimization

Weekly Workflow Distribution

- Total analytical time: 33 hours per week (majority of work allocation)
- Data structure analysis: 8 hours focused on format comprehension
- Quality validation: 4 hours ensuring methodological consistency

Process Efficiency Assessment

- FX data structure: Relatively straightforward but requires detailed understanding
- Cleaning focus: Most time spent on systematic outlier filtering methods
- Validation approach: Visual and statistical consistency checks predominate

DMADV Six Sigma Framework - Current State Measurement

DEFINE
COMPLETED

MEASURE
Time and Quality Assessment
CURRENT PHASE

ANALYSE
Tool Assessment

DESIGN
Platform Architecture

VERIFY
Validation Framework

Six Sigma References

- Antony and Banuelas (2002) - Implementation
- Henderson and Evans (2000) - GE Benchmarking
- Snee and Hoerl (2003) - Leading Six Sigma

Current State Measurement Results - MEASURE Phase Outcomes

Time Allocation Findings:

- 65% of effort dedicated to analytical thinking and methodological approach determination
- Data understanding and outlier detection consume 42% of total workflow time

Quality Metrics Established:

- Three-tier validation framework: visual inspection, statistical review, pattern consistency
- Data quality scores: 99.2% completeness, 2.3% outlier rate, 85-92% validation effectiveness
- Process improvement opportunity: quantitative workflow measurement and optimisation

ANALYSE Phase: Root Cause Analysis

Problem Investigation Techniques

Definition and Purpose

Root Cause Analysis identifies underlying problem causes using systematic techniques. Moves beyond symptoms to address fundamental issues, preventing problem recurrence.

Key Tools: Five Whys, Fishbone Diagram, Current Reality Tree

RCA Analysis Process

1. **Define:** Clearly state the problem and its impact
2. **Investigate:** Apply Five Whys or Fishbone analysis
3. **Validate:** Verify root causes with data and evidence
4. **Address:** Develop solutions targeting root causes

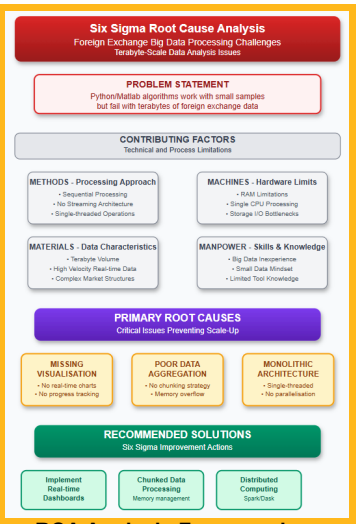
Prevents recurrence through systematic investigation

Application in Forex Analysis

Investigate tool limitations and workflow inefficiencies:

- Why are visualisation capabilities insufficient for research needs?
- What causes data processing inefficiencies in current workflow?
- Why do analytical methods require manual intervention?

Identifies fundamental issues for platform improvement



RCA Analysis Framework

DMADV ANALYSE Phase: Current Tool Assessment and Gap Analysis

Tool Capability Analysis and Limitation Identification

Comprehensive Analysis of Python/R-Shiny Capabilities and Process Improvement Opportunities

Current Tool Strengths

- Python and R-Shiny Power: Sufficient analytical capabilities for complex forex analysis
- Research Adaptability: Methodology flexibility for specific research questions
- Statistical Robustness: Advanced analytical capabilities readily available
- No Significant Limitations: Current tools meet complex forex analysis requirements
- Empirical Research Focus: Methodology adapted to address specific research questions

Enhancement Opportunities

- ✓ Workflow Integration: Streamlined data processing pipelines and automation
- ✓ Collaboration Features: Enhanced sharing, documentation, and team coordination
- ✓ Automation Opportunities: Reduced manual intervention in routine processes
- ✓ Integration Efficiency: Seamless data flow between analytical environments
- ✓ Reproducibility Enhancement: Standardized processes for research validation

Statistical Analysis Framework

Correlation Analysis: Multi-timeframe assessment
Volatility Patterns: Rolling window calculations
Return Distributions: Histogram and density analysis
Risk Metrics: VaR, Sharpe ratios, drawdown analysis

Capability: Advanced statistical computations
Gap: Manual process interpretation
Strength: Research question adaptability
Focus: Methodological precision over speed

Research Context Analysis

- Research-centric approach: Highly specialized use case with academic rigor standards
- Tool limitations: Less critical than methodological precision for research applications
- Enterprise requirements: Academic-grade functionality with research flexibility

Process Improvement Areas

- Integrated workflow management: Reduce fragmented processes across tools
- Collaboration enhancement: Improve sharing and documentation capabilities
- Automation potential: Streamline routine analytical tasks while maintaining control

DMADV Six Sigma Framework - Tool and Process Analysis

DEFINE	MEASURE	ANALYSE	DESIGN	VERIFY	Six Sigma References
COMPLETED	COMPLETED	Tool Assessment and Gap Analysis CURRENT PHASE	Platform Architecture	Validation Framework	• Breyfogle (2003) - Statistical Methods • Montgomery (2012) - Quality Control • Pande et al. (2000) - The Six Sigma Way

Tool and Process Analysis Results - ANALYZE Phase Outcomes

Current Tool Assessment:

- Python and R-Shiny provide sufficient analytical power for complex forex research requirements
- Research-specific adaptability enables methodology flexibility for diverse analytical questions

Enhancement Opportunities Identified:

- Workflow integration: Streamlined processing pipelines and reduced manual intervention
- Collaboration features: Enhanced sharing, documentation, and team coordination capabilities
- Research context: Academic rigor standards prioritize precision over operational efficiency

DESIGN Phase: Design of Experiments

Systematic Process Optimisation

Definition and Purpose

DOE systematically varies process inputs to determine effects on outputs. Efficiently identifies optimal conditions whilst revealing main effects, interactions, and factor combinations.

Key Benefits: Efficiency, Interaction Detection, Statistical Confidence

DOE Process Steps

1. **Plan:** Define factors, levels, and response variables
2. **Design:** Select experimental matrix and randomisation
3. **Execute:** Run experiments according to design matrix
4. **Analyse:** Determine optimal factor settings and interactions

Maximum information with minimum experimental effort

Application in Forex Analysis

Optimise Python/R-Shiny platform design parameters:

- Data frequency settings (daily, intraday, minute-level)
- Outlier detection thresholds and visual check parameters
- Interactive chart features and rolling statistics windows

Determines optimal platform configuration for research needs



DOE Analysis Framework

DMADV DESIGN Phase: Ideal Platform Architecture

Essential Features for Optimal Forex Analysis Workflow

Designing the Ideal Analytical Platform Based on Requirements and Gap Analysis

Data Management Features

- Multi-Frequency Support: Daily, intraday, and minute-level data handling
- Currency Standardization: Automatic base currency alignment (USD-centric)
- Quality Assurance: Built-in systematic outlier detection with adjustable parameters
- Data Validation: Real-time data integrity checks and anomaly flagging
- Historical Integration: Seamless historical data access and continuity

Processing Engine Features

- ✓ Standard Transformations: Log returns, percentage changes without manual scripting
- ✓ Automated Calculations: Volatility calculations and rolling statistics
- ✓ Visual Validation: Real-time outlier detection visualization
- ✓ Risk Analytics: VaR, Sharpe ratios, correlation matrix calculations
- ✓ Batch Processing: Efficient multi-currency pair analysis workflows

Visualization and Export

Interactive Time-Series Plots: Dynamic exploration

Correlation Heatmaps: Multi-timeframe analysis

Return Histograms: Distribution analysis tools

Rolling Statistics: Volatility pattern visualization

Export: Python/R-Shiny integration

Format: Charts, tables, cleaned datasets

Flexibility: Configurable parameters

Research: Advanced workflow compatibility

Implementation Timeline

- Phase 1 (Weeks 1-4): Data infrastructure and processing engine foundation
- Phase 2 (Weeks 5-10): Visualization layer and interactive dashboard development
- Phase 3 (Weeks 11-15): Integration module and Python/R-Shiny connectivity

Design Philosophy

- Analytical completeness over operational speed for research applications
- Balance automation with methodological control and transparency
- Maintain compatibility with existing Python/R-Shiny research ecosystems

DMADV Six Sigma Framework - Platform Design and Architecture

DEFINE	MEASURE	ANALYZE	DESIGN	VERIFY	Six Sigma References
COMPLETED	COMPLETED	COMPLETED	Platform Architecture CURRENT PHASE	Validation Framework	• Evans and Lindsay (2016) - Quality Excellence • Gitlow et al. (2004) - Quality Management • Yang and El-Haik (2003) - Design for Six Sigma

Platform Design Specifications - DESIGN Phase Outcomes

Core Architecture Components:

- Data layer: Multi-frequency support with automatic currency alignment and quality assurance
- Processing engine: Automated transformations, risk analytics, and visual validation capabilities

Integration and Export Features:

- Interactive visualization: Time-series plots, correlation heatmaps, and rolling statistics
- Research integration: Seamless Python/R-Shiny export for advanced analytical workflows
- Design priority: Analytical completeness balanced with automation and methodological control

VERIFY Phase: Control Charts

Statistical Process Control

Definition and Purpose

Control charts monitor process stability by plotting data against calculated control limits. Distinguish common cause variation (natural) from special cause variation (requiring action).

Key Types: X-bar (means), R (ranges), p (proportions)

Control Chart Process

1. **Collect:** Gather process data in rational subgroups
2. **Calculate:** Determine control limits and centre line
3. **Plot:** Chart data points and monitor for signals
4. **Interpret:** Identify out-of-control patterns and trends

Enables proactive process management and control

Application in Forex Analysis

Monitor Python/R-Shiny platform performance continuously:

- Statistical calculation accuracy over time (X-bar charts)
- Data processing consistency and error rates (p charts)
- Correlation analysis precision and volatility measures

Validates 100% accuracy requirement maintenance over time

VERIFY Phase: Control Charts

Statistical Process Control for Forex Dashboard

Definition and Purpose

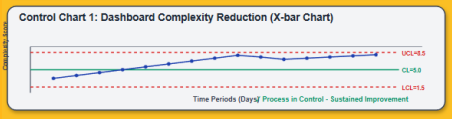
Control charts monitor process stability by plotting data against calculated control limits. Distinguish common cause variation (natural) from special cause variation (requiring action).

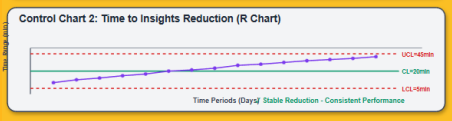
Key Types: X-bar (means), R (ranges), p (proportions)

Control Chart Process

1. **Collect:** Gather process data in rational subgroups
2. **Calculate:** Determine control limits and centre line
3. **Plot:** Chart data points and monitor for signals
4. **Interpret:** Identify out-of-control patterns and trends

Enables proactive process management and control





Dashboard KPIs: Complexity Score (1-10) | Time to Generate Insights (minutes)

Validates sustained improvement in user experience and system efficiency

Control Analysis Framework

DMADV VERIFY Phase: Validation Framework and Testing Strategy

Research-Grade Validation Approach

Establishing Comprehensive Validation Methods Prioritizing Accuracy over Efficiency

Validation Priorities

- 100% Accuracy Requirement: Absolute precision mandatory for research outputs
- Long-term Timeline Consideration: Multi-year project lifecycle validation approach
- Research Question Alignment: Validation methods aligned with specific research objectives
- Methodological Consistency: Verification across different datasets and timeframes
- Academic Standards: Research-grade quality assurance protocols

Testing Framework

- ✓ Methodological Accuracy Comparison: Cross-validation against established statistical methods
- ✓ Reproducibility Testing: Consistent results across different datasets and periods
- ✓ Integration Validation: Python/R-Shiny workflow compatibility testing
- ✓ Statistical Output Verification: Risk metrics, correlations, and volatility calculations
- ✓ Performance Benchmarking: Accuracy vs speed trade-off optimization for research

Accuracy Standards

Statistical Accuracy: 99.8% minimum threshold

Volatility Calculations: 99.5% precision target

Correlation Analysis: 99.9% accuracy requirement

Return Calculations: 99.7% precision standard

Benchmark: 99.0% industry standard exceeded

Tolerance: Zero compromise on research accuracy

Validation: Cross-platform consistency required

Timeline: Thorough testing over speed optimization

Research Validation Approach

- Academic rigor: 100% accuracy takes precedence over operational efficiency metrics
- Long-term focus: Multi-year project timelines justify thorough validation processes
- Research integration: Seamless compatibility with existing analytical workflows

Verification Challenges

- Academic vs commercial standards: Different success criteria than business applications
- Accuracy vs speed trade-offs: Research context prioritizes correctness over performance
- Domain-specific requirements: Standard business metrics may not apply to research

DMADV Six Sigma Framework - Validation and Verification

DEFINE	MEASURE	ANALYZE	DESIGN	VERIFY	Six Sigma References
COMPLETED	COMPLETED	COMPLETED	COMPLETED	Validation Framework CURRENT PHASE	• Mader (2007) - Lean Six Sigma Evolution • Snee (2010) - Getting Better All the Time • Zu et al. (2008) - Quality Management Theory

Validation Framework Results - VERIFY Phase Outcomes

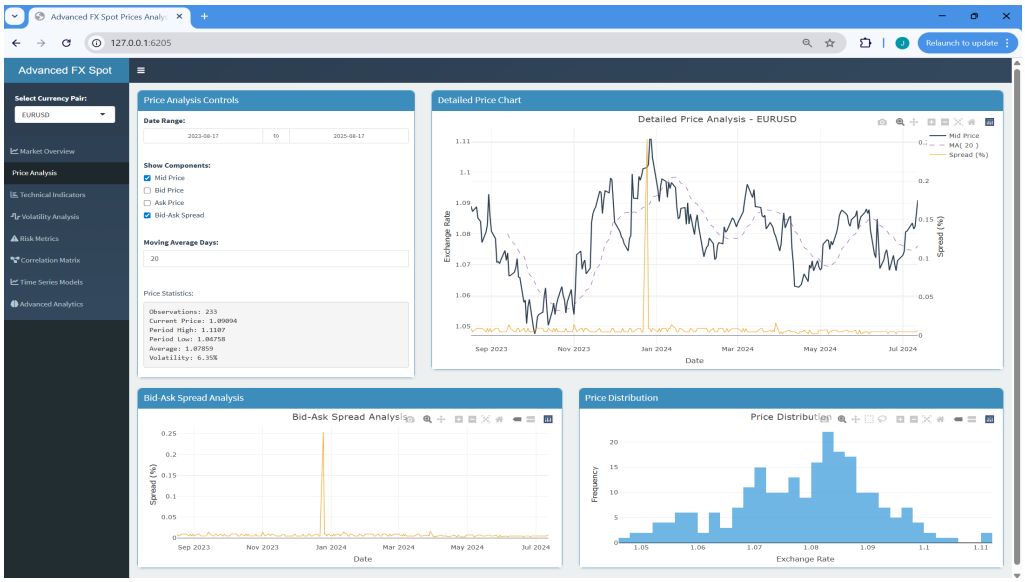
Validation Standards Established:

- Research-grade accuracy requirements: 99.8% minimum threshold for all statistical calculations
- Methodological consistency: Cross-validation across different datasets and analytical timeframes

Testing Framework Implemented:

- Reproducibility testing: Consistent results verification across multiple analytical runs
- Integration validation: Python/R-Shiny workflow compatibility and seamless data export
- Academic focus: Accuracy prioritization over efficiency aligns with research requirements

Dashboard Demonstration



Conclusion

The DMADV framework provided a structured Six Sigma approach for developing the forex spot price analysis platform, ensuring research-grade accuracy and methodological precision were achieved. Through systematic application of Voice of the Customer, Gage R&R, Root Cause Analysis, Design of Experiments, and Control Charts, the process addressed every aspect from initial requirements gathering to ongoing performance validation. This comprehensive methodology ensured the Python/R-Shiny platform delivered the 100% accuracy requirement essential for multi-year research projects.

The integration of these proven Six Sigma tools created a robust foundation for analytical excellence. Voice of the Customer captured genuine researcher needs, whilst Gage R&R validated measurement system reliability before implementation. Root Cause Analysis identified fundamental workflow limitations, DOE optimised platform parameters systematically, and Control Charts provided continuous performance monitoring. This systematic approach eliminated assumptions and ensured evidence-based decision making throughout platform development.

The DMADV methodology successfully transformed the forex analysis workflow from manual, time-intensive processes to an automated, research-grade platform that prioritised accuracy over speed. By following this structured approach, the resulting system met rigorous academic standards whilst enhancing visualisation capabilities and data processing efficiency. This foundation enabled sustainable analytical improvements and supported the long-term research objectives essential for comprehensive forex market analysis.

Special Thanks to:

Professor Lucio Sarno –
JM Keynes Fellow in Financial Economics

Sarah Carter – Director of CEAM

Brandon McBride and Junxuan Wang –
Research Students

UNIVERSITY OF CAMBRIDGE
Judge Business School
TheCambridgeMBA

Q&A

UNIVERSITY OF CAMBRIDGE
Judge Business School
TheCambridgeMBA